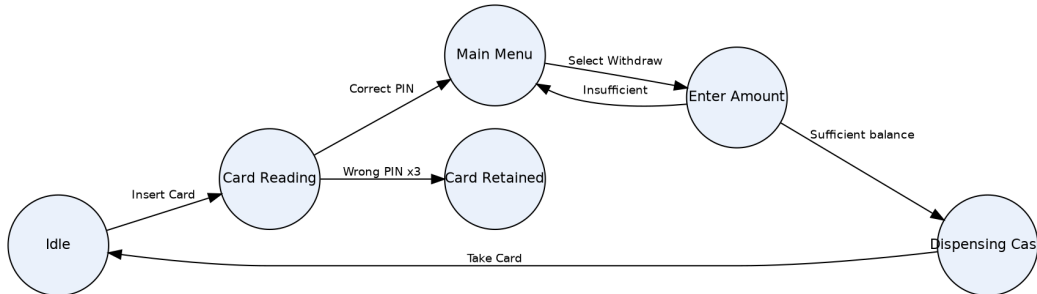


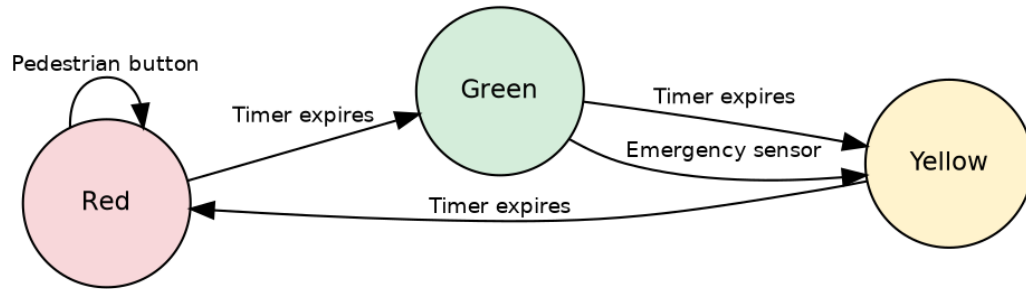
State Transition Tables

1. ATM Machine



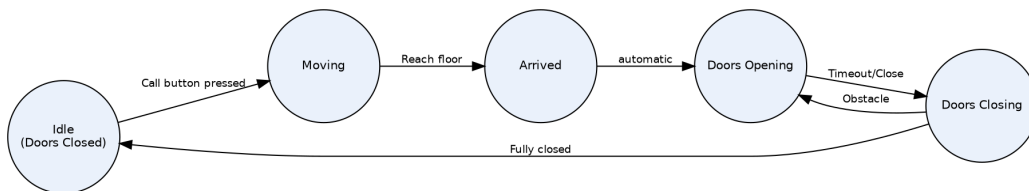
Current State	User Action	Next State	Expected Result
Idle	Insert Card	Card Reading	Card details validated
Card Reading	Enter Correct PIN	Main Menu	Access granted
Card Reading	Enter Wrong PIN (3x)	Card Retained	Card blocked/retained
Main Menu	Select Withdraw	Enter Amount	Amount input screen shown
Enter Amount	Confirm Amount (sufficient balance)	Dispensing Cash	Cash dispensed, receipt printed
Enter Amount	Confirm Amount (insufficient balance)	Main Menu	Error shown, balance unchanged

2. Traffic Light Controller



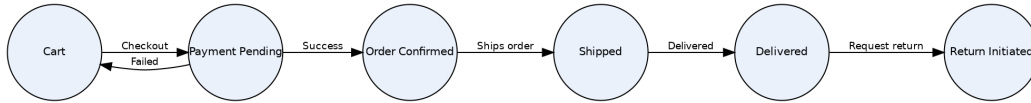
Current State	User Action	Next State	Expected Result
Red	Timer Expires	Green	Traffic allowed to move
Green	Timer Expires	Yellow	Warning before stop
Yellow	Timer Expires	Red	Traffic halted
Red	Pedestrian Button Pressed	Red (Extended)	Pedestrian crossing signal activates
Green	Emergency Vehicle Sensor Triggered	Yellow → Red	Priority given to emergency vehicle

3. Elevator System



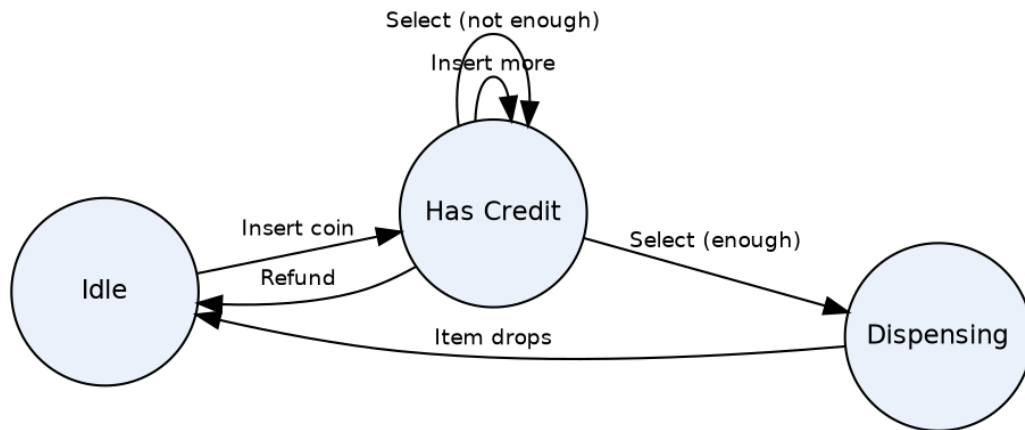
Current State	User Action	Next State	Expected Result
Idle (Doors Closed)	Call Button Pressed (Floor)	Moving	Elevator starts moving toward requested floor
Moving	Reach Target Floor	Arrived	Elevator stops at floor
Arrived	— (automatic)	Doors Opening	Doors open for boarding/exit
Doors Opening	Timeout / Close Button	Doors Closing	Doors close
Doors Closing	Obstacle Detected	Doors Opening	Doors reopen for safety
Doors Closing	Doors Fully Closed	Idle	Elevator ready for next call

4. E-commerce Order Process



Current State	User Action	Next State	Expected Result
Cart	Click Checkout	Payment Pending	Payment form shown
Payment Pending	Payment Successful	Order Confirmed	Confirmation email sent
Payment Pending	Payment Failed	Cart	Error shown, cart retained
Order Confirmed	Warehouse Ships Order	Shipped	Tracking number generated
Shipped	Package Delivered	Delivered	Order marked complete
Delivered	Customer Requests Return	Return Initiated	Return process begins

5. Vending Machine



Current State	User Action	Next State	Expected Result
Idle	Insert Coin	Has Credit	Balance updated, display shows amount
Has Credit	Insert More Coins	Has Credit	Balance increases
Has Credit	Select Item (sufficient balance)	Dispensing	Item dispensed
Has Credit	Select Item (insufficient balance)	Has Credit	"Insufficient funds" message shown
Dispensing	Item Drops	Idle	Change returned (if any), ready for next user
Has Credit	Press Refund	Idle	Coins returned